

Mohammadreza Nemati

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EXPERIENCE

Senior AI Engineer

05/2025 - Present

Volkswagen Group of America

Belmont, CA

- Fine-tuned multimodal foundation models using SFT, DPO, and GRPO for specialized planner, reasoner, grounder, verifier, and error recovery components in real-device GUI automation.
- Built multimodal GUI-agent system for real-device Android app and browser automation, combining visual screen understanding, element grounding, tool-use actions, verification, and error recovery across Android platforms.
- Developed long-horizon multimodal reasoning workflows for GUI agents, including dynamic planning, task decomposition, app/browser navigation, and completion-evidence tracking for evaluation of multi-step goals.
- Compressed VLMs using pruning, knowledge distillation, and quantization-aware training, then served custom checkpoints with vLLM continuous batching and optimized attention backends to improve GUI-agent inference throughput and latency.

Artificial Intelligence Research Assistant

08/2022 - 06/2025

Case Western Reserve University

Cleveland, Ohio

- Optimized LLM inference memory by analyzing KV-cache key/value norm imbalance, achieving 2× memory reduction with >94% accuracy retention across models up to 70B parameters [\[paper link\]](#)
- Developed LIT-LVM using latent variable embeddings, structured regularization, and GNN-based higher-order interactions, outperforming GNN, Factorization Machine, and elastic net baselines in ranking and recommendation tasks [\[paper link\]](#)
- Developed DEASurv, a deep multimodal model leveraging entity-specific embeddings, attention mechanisms, and contrastive loss to improve prediction on large-scale medical datasets

Software Engineer - AI/ML Intern

02/2024 - 07/2024

Sherwin-Williams

Cleveland, Ohio

- Designed and deployed real-time vision-transformer inference pipeline for SAM, tuning latency and improving segmentation quality by 7% IoU through inference evaluation and pipeline optimization.
- Built VLM-based visual grounding system integrating LLaVA with vision transformers, implementing cross-modal alignment for natural-language object grounding and segmentation.

Data Science Research Intern

05/2023 - 09/2023

Beyond Limits AI

Glendale, California

- Developed and deployed Graph Neural Network models (GCN, GAT, GraphSAGE) using PyTorch Geometric for graph-structured data, achieving 10% accuracy improvement over baseline approaches
- Fine-tuned Llama-2 using PyTorch distributed training with FSDP/DDP for multi-GPU optimization and quantization for efficient deployment.

SKILLS

ML Engineering: Python, PyTorch, Distributed Training (FSDP/DDP), Fine-tuning, Quantization, Knowledge Distillation, Model Evaluation

VLM / Multimodal AI: Vision-Language Models, Multimodal Reasoning, Visual Grounding, Cross-Modal Alignment, Computer Vision

Agentic Systems: GUI Agents, Tool Use, Long-Horizon Planning, Task Decomposition, Action Verification, Error Recovery

Inference Systems: vLLM, CUDA, FlashAttention-style Attention, Model Serving, GPU Memory Optimization, Latency/Throughput Tuning

EDUCATION

Case Western Reserve University – PhD, Computer Science

Cleveland, OH | 2025

University of Toledo – MS, Computer Science

Toledo, OH | 2020

Shiraz University – BS, Electrical Engineering

Shiraz, Iran | 2017